



Department of Mechanical Engineering

Dear NSF Selection Committee,

It is my honor to recommend Mr. Jacob Vaught for the NSF Graduate Research Fellowship. I am a Professor of Mechanical Engineering at the University of South Carolina and the Principal Investigator of the Integrated Multiphysics and Systems Engineering Laboratory (iMSEL), where Jacob has conducted research under my supervision. In my 15+ years in academia, I have worked with dozens of bright students, and Jacob ranks among the very best (top X%) I have had the pleasure to advise. He is an outstanding young researcher with a rare combination of technical prowess, leadership ability, and vision for how engineering innovations can benefit society. I offer my strongest possible endorsement of his application.

I have supervised Jacob's work since his junior year, during which he has grown from a promising student into a key contributor and leader in my research group. Jacob's primary project in our lab centers on the integration of radar and camera systems for detecting small maritime targets – an ambitious undertaking that spans multiple engineering and computing disciplines. This project is directly aligned with the research plan Jacob has outlined for the GRFP, and I am delighted to say that he has already made impressive progress on it.

In our lab's multi-sensor research initiative, Jacob quickly emerged as a technical leader. When a discrepancy arose between the radar data format and the camera data processing pipeline, Jacob aimed to resolve it – he wrote a conversion script and established a standardized data format that both radar teams now use. Moreover, he has been a go-to person for troubleshooting; whether it's fine-tuning the radar's hardware settings or improving the computer vision algorithm's performance, Jacob dives in with a systematic approach. His ability to collaborate across disciplines – from discussing antenna characteristics with electrical engineers to refining neural network parameters with computer scientists – is highly beneficial to our work. It reflects his intellectual versatility and superb communication skills.

As the lab's Principal Investigator, I have been delighted to see Jacob take on a mentorship role. He has directly supervised two newer undergraduate researchers who joined our project this past year. Jacob trained them in operating the radar equipment safely, in processing the collected data, and in using the software tools he developed. He did so with patience and professionalism, holding weekly mini-training sessions and preparing example datasets for them to practice with. Thanks to Jacob's guidance, these students have become productive team members much faster than I anticipated. This

mentorship not only showcases Jacob's leadership and empathy, but it also multiplies the broader impact of his skills by transferring knowledge to the next cohort of students. In effect, Jacob has been an unofficial project manager for our radar detection team – and he excels at it.

What truly sets Jacob apart is his holistic understanding of research impact. He is not just solving technical problems in isolation; he is keenly aware of the broader context and potential benefits of his work. In terms of Intellectual Merit, Jacob's contributions demonstrate creativity, depth, and the ability to integrate knowledge from different fields. He is often asking the kind of probing questions that lead to new insights.

Equally important are Jacob's Broader Impacts, which are already evident in his activities. A cornerstone of Jacob's proposed graduate work is the creation of a comprehensive open dataset of labelled radar data for small object detection on water. Jacob has been the driving force behind this initiative. He drafted a detailed data collection and annotation plan, including the deployment of sensors at a local lake to gather diverse scenarios (day/night, various weather conditions). He has advocated from the start that this dataset be made publicly available to other researchers. In doing so, Jacob is directly contributing to the broader scientific community – such a dataset will enable advancements in maritime sensing well beyond the scope of our lab.

Jacob also engages in more traditional outreach and broader impact efforts. He has volunteered with our college's K-12 outreach programs, bringing engineering demos to local schools. It was a joy to watch him interact with the kids; he was approachable, enthusiastic, and able to explain complex concepts in a fun, relatable way. This speaks to Jacob's ability to inspire the next generation of STEM students, and it highlights his commitment to sharing the excitement of scientific discovery with a broader audience. Additionally, within the university, Jacob has been active in the ASME and the IEEE student chapters, where he's organized multiple technical workshops. These efforts show that Jacob is a disseminator of knowledge – a quality that will make him a well-rounded graduate student and future educator.

Jacob is a once-in-a-decade talent in my experience – he has the technical skills of a graduate student, the leadership qualities of a seasoned engineer, and the humanitarian perspective of a person who wants his work to benefit others. I have every confidence that Jacob will continue to thrive in graduate school and make important contributions to the field of intelligent sensor systems. He has my highest recommendation for the NSF Graduate Research Fellowship. I am excited to see all that he will accomplish with the support of this fellowship, and I believe he will be a credit to the program.

Sincerely,
Dr. Yi Wang
Professor of Mechanical Engineering (Frank B. Herty Chair)
Principal Investigator, iMSEL Research Group
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